

Explainer: Problem 2 (Rudin 7.7) — Uniform Convergence vs. Derivative Interchange

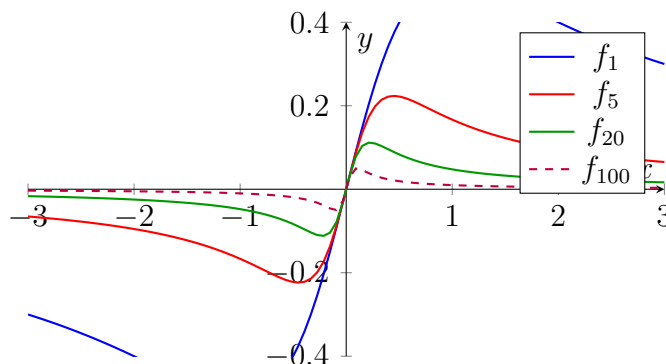
What's going on?

We have $f_n(x) = \frac{x}{1+nx^2}$. This sequence converges to $f(x) = 0$ for all x . The question asks:

- Is the convergence uniform? (Yes.)
- Can we swap limit and derivative: $f'(x) = \lim f'_n(x)$? (Only for $x \neq 0$.)

This is a cautionary tale: **uniform convergence of f_n does NOT guarantee convergence of f'_n .**

Visualizing f_n



Each f_n has a bump near the origin that gets **taller and narrower** as n grows... but not taller than a bound that $\rightarrow 0$. The maximum of $|f_n|$ is $\frac{1}{2\sqrt{n}}$ (achieved at $x = \pm 1/\sqrt{n}$), so the sup-norm $\|f_n\|_\infty \rightarrow 0$.

Why uniform convergence holds

To show $f_n \rightarrow 0$ uniformly, we need $\sup_x |f_n(x)| \rightarrow 0$.

Finding the maximum of $|f_n|$: By AM-GM or calculus, for $x > 0$:

$$f_n(x) = \frac{x}{1+nx^2} = \frac{1}{1/x + nx} \leq \frac{1}{2\sqrt{n}}$$

since $1/x + nx \geq 2\sqrt{n}$ by AM–GM. Equality when $x = 1/\sqrt{n}$.

So $\|f_n\|_\infty = \frac{1}{2\sqrt{n}} \rightarrow 0$. Uniform convergence!

The derivative story

Now compute $f'_n(x) = \frac{1-nx^2}{(1+nx^2)^2}$.

At $x \neq 0$: As $n \rightarrow \infty$, both numerator and denominator are dominated by the nx^2 terms, so $f'_n(x) \rightarrow 0$. Since $f(x) = 0$, we have $f'(x) = 0 = \lim f'_n(x)$. ✓

At $x = 0$: $f'_n(0) = \frac{1-0}{(1+0)^2} = 1$ for every n . So $\lim f'_n(0) = 1$. But $f(x) = 0$, so $f'(0) = 0 \neq 1$. 55

The lesson

$f_n \rightarrow f$ uniformly does
NOT imply $f'_n \rightarrow f'$
(Theorem 7.17 requires uniform convergence of the *derivatives* f'_n , not just the functions.)

The slopes at $x = 0$ stay at 1 forever even though the functions flatten to 0 everywhere. The bump gets narrow so fast that its height vanishes, but its slope at the origin never changes.